

RESEARCH STATEMENT

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1. INTRODUCTION

1.1. Summary

I work in algebraic combinatorics, using the combinatorics of reflection groups to study algebraic varieties and their coordinates. My main tool is the *affine Deodhar decomposition*: a partition into explicit pieces of the form $(\mathbb{C}^*)^r \times \mathbb{C}^s$, where $\mathbb{C}^* = \mathbb{C} \setminus \{0\}$. These pieces are products of algebraic tori and affine spaces, indexed by periodic crossing-and-elbow diagrams that I call *affine Deograms* [17]. I keep coming back to this decomposition because it can be read in two ways.

Read as an enumeration, the decomposition produces families counted by rational Catalan numbers. I construct bijections from maximal Deograms to lattice paths above a diagonal and their labeled versions, the rational Dyck paths and parking functions. Read as a source of coordinates, the diagrams encode candidate cluster seeds: systems of coordinate functions related by prescribed exchange rules. With Moriah Elkin and Matthew J. Tyler, I use them to study varieties in affine type A . Both programs use the same diagrammatic framework: one studies what the pieces enumerate, while the other uses them as coordinate charts.

1.2. Background and motivation

The *Grassmannian* $\mathrm{Gr}(k, n)$ is the space of k -dimensional subspaces of $\mathbb{C}^n$, represented by rank- k matrices modulo row operations. Its *Plücker coordinates* Δ_I are the maximal minors indexed by k -subsets $I \subset [n] = \{1, \dots, n\}$. An *open positroid variety* Π_f° is the locus where the ranks of all cyclically consecutive collections of columns take prescribed values [15, 24]. These values are encoded by a *bounded affine permutation*: a bijection $f: \mathbb{Z} \rightarrow \mathbb{Z}$ satisfying $f(i+n) = f(i) + n$, $i \leq f(i) \leq i+n$, and $\sum_{i=1}^n (f(i) - i) = kn$.

Galashin and Lam associate to Π_f° a *positroid Catalan number* C_f : remove the factor $(q-1)^{n-c(f)}$ from its finite-field point-count polynomial and then set $q=1$, where $c(f)$ counts the cycles of f modulo n [10]. When $c(f)=1$, this is the Euler characteristic $\chi(\Pi_f^\circ/T)$, the alternating sum of Betti numbers after quotienting by the diagonal torus T , which rescales the ambient coordinates. For the top cell $f_{k,n}: i \mapsto i+k$ with $\gcd(k, n)=1$, it is the rational Catalan number

$$C_{f_{k,n}} = \mathrm{Cat}_{k,n-k} = \frac{1}{n} \binom{n}{k},$$

and the cohomology of Π_f° , with its weight and Hodge gradings, produces q, t -Catalan refinements [11]. Deodhar's decomposition indexes the torus and affine-space pieces by *distinguished subexpressions*: choices of factors in a word of simple reflections, with every step that decreases the length of the running product required [6, 16]. It expresses C_f as the number of *maximal Deograms*, the diagrams with the fewest elbows; their strata are precisely the terms that survive the normalized $q=1$ count. Galashin–Lam–Trinh–Williams used these objects and their labeled counterparts to define rational noncrossing Coxeter–Catalan objects and parking objects in every finite Weyl group [13]. In type A , the noncrossing family consists of maximal Deograms, while the classical nonnesting family consists of rational Dyck paths and parking functions [1]. A bijection would let us transfer statistics, cyclic actions, and labels between these families, and that is what I set out to build.

The same decomposition has a geometric reading. *Plabic graphs*, planar graphs in a disk with black and white internal vertices, parametrize the positive part of Π_f° , where its nonzero Plücker coordinates are positive [8, 9, 24]. They also produce cluster seeds. A *seed* consists of algebraically independent coordinate functions and a *quiver*, a directed graph encoding their

exchange relations. *Mutation* replaces one mutable function using an exchange relation and changes the quiver; frozen functions remain fixed. The corresponding coordinate charts are algebraic tori. In finite type, 3D plabic graphs extend this construction to *braid varieties*, spaces of sequences of flags—nested chains of subspaces—with relative positions prescribed by a braid word [4, 12]. Ordinary plabic graphs and *Le-diagrams*, fillings of shapes inside a rectangle encoding positive Deodhar strata, give a finite model. I ask what replaces them in an arbitrary Plücker chart $\Delta_I \neq 0$ and in the affine setting, where periodic diagrams are forced.

2. RATIONAL COXETER–CATALAN COMBINATORICS

2.1. From maximal Deograms to Dyck paths

An f -Deogram is a periodic crossing-and-elbow diagram whose chosen crossings form a distinguished subexpression for f ; at an elbow, two strands turn rather than cross. It is *maximal* if it has the minimum possible number of elbows. Write $\text{Deo}_f^{\max}$ for this set and $\text{Dyck}_{k,n}$ for the *rational Dyck paths*: north-and-east lattice paths from $(0, 0)$ to $(n - k, k)$ that remain above the joining diagonal. Galashin and Lam asked for a bijection explaining $\# \text{Deo}_f^{\max} = C_f$ object by object [10, Problem 7.6]; my paper in preparation [18] gives one.

Theorem 1 (Martinez [18]). *Let f be a bounded affine permutation. There is a recursively defined bijection from $\text{Deo}_f^{\max}$ to the set of Dyck paths avoiding the region $\Gamma(f)$ determined by f . For the top cell $f = f_{k,n}$ with $\gcd(k, n) = 1$ this is a bijection*

$$\Phi_{k,n} : \text{Deo}_{f_{k,n}}^{\max} \longrightarrow \text{Dyck}_{k,n}.$$

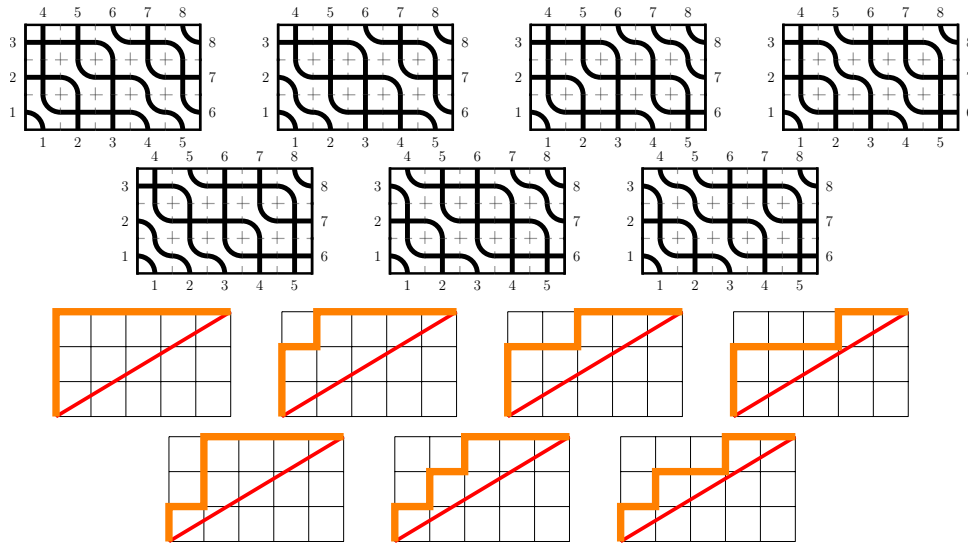


Figure 1: Maximal $f_{3,8}$ -Deograms (top) and rational Dyck paths (bottom), illustrating the two sides of Theorem 1. From [17].

The proof matches a reversible reduction of the periodic strand diagram with a decomposition of the Dyck path into smaller rational pieces. Its local choices produce both $\Phi_{k,n}$ and its inverse. The reduction step is a change of Plücker chart carried out with the affine-diagram moves of [17], and in the process it resolves the consecutive-chart problem posed there [17, Problem 7.1].

2.2. Direct bijections and parking functions in the Fuss–Catalan case

In the *Fuss–Catalan* specialization $n = dk + 1$, with d a positive integer, periodicity makes a direct map possible. A *rational parking function* is a rational Dyck path whose k north steps are labeled by $1, \dots, k$, increasingly within each column. Write $\text{PF}_{k,n}$ for this set and $\text{Deo}_{f_{k,n}}^{\text{tw}}$ for the twisted maximal Deograms, the parking counterparts in the noncrossing model of [13].

Theorem 2 (Martinez [19]). *Assume $0 < k < n$ and $n \equiv 1 \pmod{k}$. There is a bijection*

$$\tilde{\Psi}_{k,n} : \text{Deo}_{f_{k,n}}^{\text{tw}} \longrightarrow \text{PF}_{k,n}.$$

This realizes the ordinary and parking theories of [13] in one framework [1, 21]. The geometry is present in the count itself: in [19] I show that the parking-function number arises as a sum of Euler characteristics of torus quotients of open positroid varieties,

$$\sum_{u \in S_a} \chi(\Pi_{f_u}^\circ / T) = b^{a-1},$$

where S_a is the group of permutations of a labels and f_u indexes the open positroid variety $\Pi_{f_u}^\circ \subset \text{Gr}(a, a+b)$ associated to u in the parking construction. Thus parking objects, like Catalan objects, are indexed by maximal Deograms in the affine Deodhar decomposition.

Two Dyck-path statistics drive the next step: *area* counts unit squares wholly between the path and diagonal, while *dinv* counts pairs of an east step and a later north step intersected by a common line parallel to the diagonal [11]. I plan to identify their counterparts on maximal Deograms, so that the two-variable refinement $C_{k,n}(q, t)$ becomes a weighted sum over those diagrams; extend the direct construction beyond Fuss slope; and test compatibility with cyclic actions and local flips. These refinements would connect Deograms to cyclic sieving, where polynomial evaluations at roots of unity count fixed points, and to Tamari-type orders generated by directed local moves.

3. AFFINE 3D PLABIC GRAPHS AND CLUSTER STRUCTURES IN AFFINE TYPE A

3.1. From Plücker charts to the affine flag variety

Let $I \in \binom{[n]}{k}$ and consider the affine patch

$$U_{f,I} := \Pi_f^\circ \cap \{\Delta_I \neq 0\}.$$

Snider identified $U_{f,I}$ with an *open Richardson variety* $\hat{R}_{f,t_I}$, the intersection of a Schubert cell and an opposite Schubert cell, or equivalently of orbits of opposite triangular subgroups [25]. Here $t_I(i) = i + n$ when $i \bmod n \in I$, and $t_I(i) = i$ otherwise. The ambient *affine flag variety* parametrizes periodic chains of lattices in $\mathbb{C}((z))^n$; a lattice is a free rank- n $\mathbb{C}[[z]]$ -module spanning that Laurent-series vector space. It is attached to the affine Kac–Moody group $\widehat{\text{SL}}_n$, the loop-group analogue of SL_n . Thus questions about a Plücker chart become questions in affine flag geometry. More generally, *double affine braid varieties* $\hat{R}_{f,\beta}$ are affine flag-configuration spaces indexed by f and a positive double affine braid word β . They extend affine Richardson varieties and the finite-type braid varieties whose cluster structures were found by Galashin–Lam–Sherman–Bennett–Speyer and Casals–Gorsky–Gorsky–Le–Shen–Simental [4, 12].

Here the finite and affine settings part ways. The *cluster algebra* $\mathcal{A}$ is generated by all variables obtained by mutation, together with frozen variables and their inverses. The *upper cluster algebra* $\mathcal{U}$ is the intersection of the Laurent polynomial rings of all seeds: its functions are Laurent in every cluster chart. In finite type, braid varieties, open Richardson varieties, and open positroid varieties have coordinate rings equal to $\mathcal{A} = \mathcal{U}$ [4, 9, 12]. In Kac–Moody type the picture is incomplete. Bao and Ye proved that the coordinate rings of open Richardson varieties of symmetrizable Kac–Moody type, and of generalized Richardson varieties including braid varieties, are upper cluster algebras [2], extending Williams’s result for double Bruhat cells, intersections of Bruhat cells for opposite triangular subgroups [26]. They conjecture equality with the cluster algebra after inverting frozen variables. Whether $\mathcal{A} = \mathcal{U}$ holds—whether every function Laurent in every seed is generated by cluster variables—remains open in general beyond finite type. The finite-type proofs rely on a combinatorial model of seeds that the Kac–Moody setting lacks. With Elkin and Tyler [7], I am trying to supply that model in affine type A and close the gap.

3.2. Affine Deograms and torus patches

Let P_I be the north-and-east lattice path with n steps whose north steps are indexed by I . In [17] I introduced (f, P_I) -*affine Deograms*: periodic crossing-and-elbow diagrams whose strands encode f and whose fundamental region is cut out by P_I . The diagrams retain Deodhar’s distinguished-subexpression condition while allowing the usual rectangle to wrap around its boundary, and they index the Billig–Dyer affine Deodhar decomposition [3] of $U_{f,I}$ into products of algebraic tori and affine spaces. Here f specifies the variety and P_I the chart, so finite-field point counts depend combinatorially on I ; the local moves of [17] give correspondences preserving the point-count contribution of each stratum between geometrically isomorphic patches.

The intrinsic geometry of a patch can depend sharply on I . The top open positroid variety in $\text{Gr}(2, 4)$ has dimension four and generic diagonal-torus orbits of dimension three, yet its Δ_{13} -patch is isomorphic to $(\mathbb{C}^*)^4$ [20]. This is an “unexpected torus” in the sense of Gorsky–Kim–Sherman–Bennett, who characterized the open Richardson varieties in finite type that are algebraic tori [14]. The example shows why being an algebraic torus must be distinguished from having a dense orbit under the standard diagonal torus. With Tyler, I proved the affine analogue of their criterion, which applies to the affine Richardson varieties $\hat{R}_{f,t_I}$ and hence to every Plücker patch.

The criterion uses *Bruhat order*: $v \leq w$ if a reduced word in simple reflections for w contains such a word for v as a subword. Write $\ell(w)$ for the length of a reduced word in simple reflections. The *Kazhdan–Lusztig R-polynomial* $R_{v,w}(q)$ counts points of the corresponding open Richardson variety over a field with q elements.

Theorem 3 (Martinez–Tyler [20]). *Let $f \leq t_I$ and set $d = \ell(t_I) - \ell(f)$. The following conditions are equivalent:*

- (i) $U_{f,I} \cong (\mathbb{C}^*)^d$ as an algebraic variety;
- (ii) the affine Bruhat interval $[f, t_I]$ contains no 2-crown, a subinterval isomorphic to the six-element Bruhat order on S_3 ;
- (iii) the Kazhdan–Lusztig R-polynomial satisfies $R_{f,t_I}(q) = (q - 1)^d$;
- (iv) the affine Deodhar decomposition has one component, equivalently there is a unique (f, P_I) -affine Deogram.

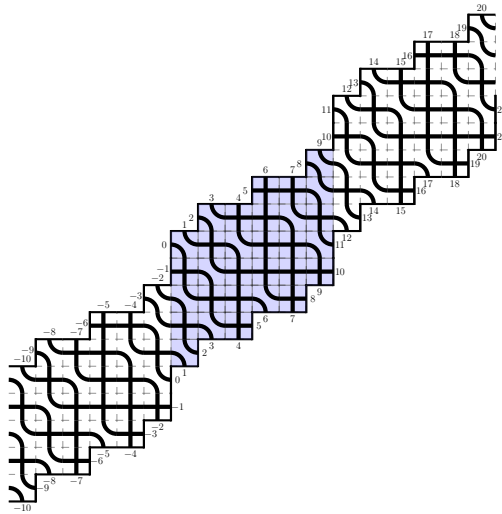


Figure 2: An (f, P) -affine Deogram. The shaded block is a representative region of the periodic filling. From [17].

Conditions (i)–(iii) are the affine counterparts of the finite-type criterion of [14]. The theorem

identifies patches admitting a single Deodhar torus chart, giving a natural base case for the cluster program: the remaining task is to realize these torus coordinates as a candidate seed.

3.3. Affine 3D plabic graphs and the cluster conjecture

In [7] we define *affine 3D plabic graphs*: periodic spatial graphs with black and white vertices attached to a double affine braid word β . Their local moves generalize the square move, which switches the colors around a quadrilateral face, and the braid relations of finite type. Their reduced positive members are exactly the minimal affine Deograms, or *affine Le-diagrams*, of [17]. To a graph G we attach a candidate seed $\Sigma(G)$. Its variables are *generalized minors*, matrix coefficients between extremal weight vectors that extend ordinary matrix minors to $\widehat{\mathrm{SL}}_n$. We realize the minors at the affine node on the *semi-infinite wedge*, an exterior-product representation with infinitely many factors and a fixed eventual tail. Its quiver records the exchanges induced by the local moves.

A seed is *really full rank* if the rows of its extended exchange matrix span $\mathbb{Z}^m$, where m is the number of mutable variables. *Local acyclicity* means that the spectrum of its cluster algebra has a finite cover by cluster localizations whose quivers have no directed cycles among mutable vertices.

Theorem 4 (Elkin–Martinez–Tyler [7]). *Let G be an affine 3D plabic graph for $\mathring{R}_{f,\beta}$. Then the candidate seed $\Sigma(G)$ is really full rank, its cluster algebra is locally acyclic, and the local moves act on candidate seeds by mutation and quasi-cluster equivalence.*

Here *quasi-cluster equivalence* allows variables to be rescaled by monomials in frozen variables, compatibly with mutation. Local acyclicity is proved by recursively reducing the periodic quiver at sinks, vertices with no outgoing arrows, and Muller’s theory then delivers $\mathcal{A}(\Sigma(G)) = \mathcal{U}(\Sigma(G))$ together with finite generation and normality [22, 23]. Bao and Ye supply the other half, $\mathcal{U} = \mathbb{C}[\mathring{R}_{f,\beta}]$, for their seed; here $\mathbb{C}[\mathring{R}_{f,\beta}]$ denotes the ring of regular functions on the variety. What remains is to match the two seeds.

Conjecture 5 (Elkin–Martinez–Tyler [7]). *For every double affine braid variety $\mathring{R}_{f,\beta}$ and every affine 3D plabic graph G for it,*

$$\mathbb{C}[\mathring{R}_{f,\beta}] = \mathcal{A}(\Sigma(G)) = \mathcal{U}(\Sigma(G)).$$

If G and G' represent the same variety, then $\Sigma(G)$ and $\Sigma(G')$ are mutation-equivalent, with mutations realized by the local moves.

Specialized to β a word for t_I , the conjecture says that every Plücker chart $U_{f,I}$ of every positroid variety is a cluster variety with seeds indexed by affine Le-diagrams. In this case the affine 3D plabic graphs flatten to *punctured plabic graphs*, bicolored graphs drawn on a disk with one interior point removed; the puncture records the chart. The periodic combinatorics of [17] then matches the plabic combinatorics exactly.

Theorem 6 (Elkin–Martinez–Tyler [7]). *Affine Le-diagrams for the patch $U_{f,I}$ are in bijection with equivalence classes of reduced punctured plabic graphs for (f, I) under the plabic moves.*

An affine Le-diagram thus names a move-equivalence class of punctured plabic graphs; their face labels give the candidate seed’s Plücker-type functions. The patch conjecture parallels the Galashin–Lam and Fraser–Sherman–Bennett theorems for Π_f° [8, 9], with the puncture recording the chart.

4. UNDERGRADUATE RESEARCH

I first did research as an undergraduate at Harvey Mudd College, through a directed reading, two REUs, and a senior thesis, and those projects turned into seven published papers as an

undergraduate. Those projects showed me what research mathematics actually is, and that I wanted to keep doing it. My mentors' guidance is what pointed me toward graduate school, and much of how I work now—how I pick a problem, when to trust a computation, how to write a paper—I learned from them. Mentoring undergraduates is how I pass that on. Since 2023 I have mentored eight undergraduates in semester-long projects through the UCLA Directed Reading Program, on topics from the probabilistic method to Schubert polynomials, and I now serve on the program's organizing committee. In Spring 2026, Olha Shevchenko and I co-mentored three of these students on a research project that became a preprint on q -Fibonomial coefficients, the Fibonacci analogues of Gaussian binomial polynomials [5]. We proved unimodality in small cases: the coefficients increase weakly to a peak and then decrease weakly. I intend to keep mentoring projects like this.

5. FUTURE RESEARCH PROGRAM

Beyond the two programs above, the project I expect to occupy me longest is a systematic transfer of what is known for positroid varieties to their Plücker patches. Each structure attached to Π_f° should have a periodic analogue attached to $U_{f,I}$, and the analogue should depend on I in a way that the original cannot see. The dictionary so far reads: Deograms become affine Deograms [17]; Le-diagrams and plabic graphs become affine Le-diagrams and punctured plabic graphs [7]; the cluster structure of [9] becomes Conjecture 5; the toric positroid varieties become the torus patches of Theorem 3.

The entry I most want to add next is the link. Galashin and Lam attached to Π_f° a *positroid link*, a union of circles in S^3 obtained by closing a braid determined by f . Its *Khovanov–Rozansky homology* is a triply graded link invariant whose graded pieces recover the point count and q, t -Catalan number of Π_f° [11]. Finite-type braid varieties likewise have *braid Richardson links*, obtained by closing the defining positive braid followed by the inverse positive braid lift of the endpoint permutation; their link invariants encode point-count data [12]. I want an affine analogue attached to $\check{R}_{f,\beta}$, hence to every patch $U_{f,I}$, whose invariants recover the affine Deodhar point count. I would then test which torus patches can be detected by *sliceness*, bounding disjoint smooth disks in the four-ball, or by being an *unlink*, bounding disjoint disks in S^3 . The finite-type examples I have computed with Tyler show why any criterion must distinguish being an algebraic torus from having a dense standard-torus orbit [20].

I also plan to define *patch Catalan numbers* $C_{f,I}$ by counting appropriate maximal Deograms, develop their q, t -refinements, and characterize a positive real locus on each patch by positivity of the variables in one seed. At the top cell, these diagrams give the noncrossing objects of Section 2.

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